

# Unit - IV Security in Cloud Computing

Weightage marks

14

# Syllabus

4

TLO 4.1 Explain security and related risks in Cloud Computing.  
TLO 4.2 Explain key features of Data Security.  
TLO 4.3 Write steps to implement Cloud Data Security.  
TLO 4.4 Explain identity management and access facility of given Cloud set-up.  
TLO 4.5 Explain the features of Security-As-A-Cloud Service.

## **Unit - IV Security in Cloud Computing**

4.1 Cloud Security Concepts: Multi-tenancy, Virtualization, Data Outsourcing and Trust Management, Metadata security  
4.2 Cloud Risk: Concept, Types of Cloud Risks  
4.2.1 Policy and Organizational Risks  
4.2.2 Technical Risks  
4.2.3 Legal Risks  
4.3 Data security technologies, Data Security risks  
4.4 Digital Identity and Access Management  
4.5 Content level security: Pros and Cons, Features of Security-As-A-Cloud Service

# Security in Cloud Computing

- Cloud security involves protecting data, applications, and infrastructure in cloud environments from cyber threats

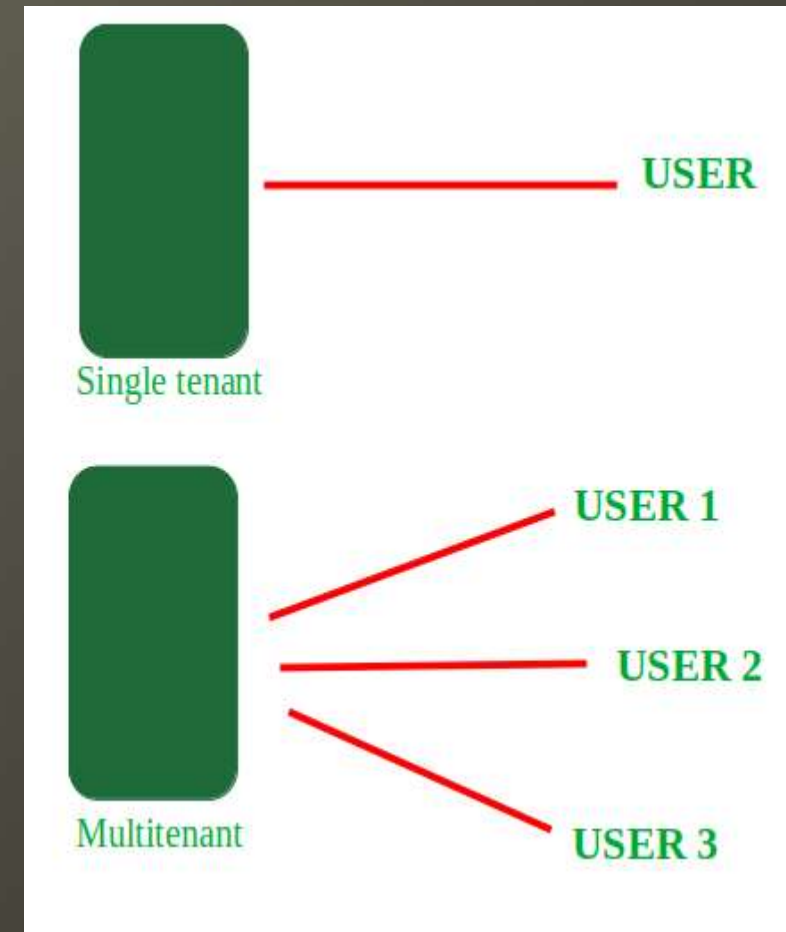
Or

- Cloud security means protecting data, applications, and services stored on the cloud from threats like hacking, data loss, and misuse. It uses tools like encryption, firewalls, and secure login to keep information safe.

# 4.1 Cloud Security Concepts (Anyone for 2m)

## 1 ) Multitenancy

- is a cloud architecture where one software or system serves multiple users (called tenants). In this model, many customers share the same computing resources (like servers or storage), but their **data remains completely separate and secure**.
- It's like living in an apartment building — everyone shares the same structure, but each person has their own private room.
- Multitenancy is also called **shared hosting**, where cloud resources are divided among different customers.
- Example -like a bank serving many customers with separate accounts.



# 4.1 Cloud Security Concepts

## 2) Virtualization

- is the process of creating virtual versions of hardware or systems, allowing multiple virtual machines to run on a single physical device.
- Virtualization is a technology that allows multiple virtual instances of operating systems, servers, storage, or networks to run on a single physical machine, maximizing resource utilization and efficiency
- **Example:** Running multiple operating systems like Windows and Linux on one computer using software like VMware is virtualization.



VM1

VM2

App App App

App App App

Operating System

Operating System



VMware Virtualization Layer

Hardware



## 4.1 Cloud Security Concepts

### 3)Data Outsourcing

- The data that powers these services and resources are hosted on remote data servers, which are deployed, monitored, and maintained by cloud outsourcing providers.
- Data outsourcing means storing or processing your data on external cloud provider systems instead of your own.
- **Example:** A company storing customer records on Google Cloud instead of its own servers.





## 4.1 Cloud Security Concepts

### 4)Trust Management

- Trust management in cloud computing ensures users can rely on the cloud provider to handle data securely and follow privacy rules.
- **Example:** A hospital uses Microsoft Azure to store patient records because it ensures end-to-end encryption, role-based access, and compliance with healthcare regulations like HIPAA – building trust in data security and privacy.

## 4.1 Cloud Security Concepts

### 5)Metadata Security

- Metadata security protects information *about* data (like file owner, creation time) from unauthorized access or leaks.\
- **Example:** If someone sees a photo's metadata showing the time and location it was taken, they might know where you were — even without seeing the photo.



## 4.2 Cloud Risk: Concept, Types of Cloud Risks

- 4.2.1 Policy and Organizational Risks
- 4.2.2 Technical Risks
- 4.2.3 Legal Risks

# Questions

2. Attempt any **THREE** of the following :

12

(d) State any four cloud security risk.

4 m

5. Attempt any **TWO** of the following :

12

(c) Explain cloud risk in detail.

6 m

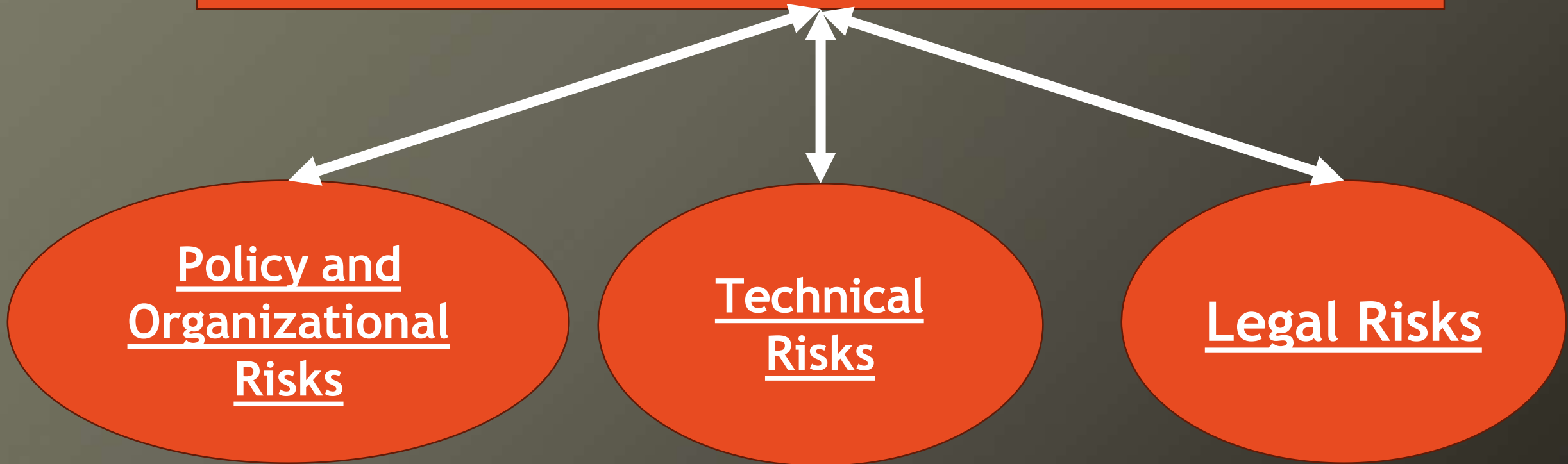


## 4.2 Cloud Risk: Concept

- Cloud risk refers to the possible problems or threats that users or organizations may face while using cloud services. Since data and applications are stored on remote servers managed by third-party providers.
- **In simpler words** , Cloud risk means the possible dangers or problems that can happen when using cloud services. These risks can affect **data security, service availability, legal compliance, and business operations**.
- **Example:** If a cloud provider goes down suddenly, your website or app may become unavailable — that's a cloud risk.

# Types of Cloud Risks

## Types of Cloud Risks



## 4.2.1 Policy and Organizational Risks

- Policy and organizational risks are the risks that arise due to lack of proper policies, weak management, poor planning, or untrained staff when using cloud computing services.
- These risks are related to how the organization manages its cloud usage, access control, staff training, and internal policies. If proper procedures are not followed, it may lead to data loss, misuse of services, or legal problems.
- **Example**
- If employees use personal cloud storage (like Google Drive) to store office data without permission, it becomes a policy risk and may lead to data leaks

# Examples of Policy and Organizational Risks:

- **Lack of Cloud Usage Policy:**
- **Weak Access Management:**
- **No Disaster Recovery or Backup Plan:**
- **Untrained Staff:**
- **Use of Unauthorized Cloud Services (Shadow IT):**

Policy and organizational risks can be reduced by creating strict cloud usage policies, controlling user access, training staff, and setting up backup and recovery systems.

## 4.2.2 Technical Risks

- Technical risks in cloud computing are risks that arise due to failures or weaknesses in hardware, software, networks, or technical configurations used in cloud services.
- These risks are mostly related to the technical infrastructure of the cloud, such as data loss, system downtime, or security flaws. If not properly managed, these can affect service availability, data privacy, and overall system performance.
- Technical risks can be minimized by using strong encryption, securing APIs, regular backups, system monitoring, and choosing reliable cloud providers.
- **Example:** If a company's cloud server crashes due to a software bug, employees can't access their files — this is a technical risk.



## Examples of Technical Risks:

- Data Breaches:
- System Downtime / Service Unavailability:
- Insecure APIs (Application Programming Interfaces):
- Data Loss or Corruption:
- Multi-tenancy Risks:

## 4.2.3 Legal Risks

- Legal risks in cloud computing refer to the risks related to laws, regulations, data privacy, and legal responsibilities when storing or processing data on the cloud.
- Cloud service users must follow national and international laws regarding data security, privacy, and ownership. If data is stored in a different country or the cloud provider fails to meet legal standards, it may lead to legal action or penalties.
- Legal risks can be reduced by reading contracts carefully, choosing trusted cloud providers, and ensuring compliance with data protection laws.
- **Example -A company storing customer data in another country without following privacy laws may face a legal penalty — this is a legal risk.**

# Examples of Legal Risks:

- Data Location Issues:
- Non-Compliance with Regulations
- Lack of Clarity in Contracts:
- Unauthorized Data Sharing:
- Jurisdiction Conflicts:

# Possible Question in Exam: (Homework!)

- Q: Differentiate between Policy & Organizational Risks, Technical Risks, and Legal Risks in Cloud Computing.

## 4.3 Data Security Technologies and Risks

- **Data Security Technologies**: These are tools and techniques used to **protect data** in the cloud from unauthorized access, loss, or corruption.
- **Common Data Security Technologies**:
  1. **Encryption**:
    - Converts data into unreadable format. And Only authorized users with a key can access it.
  2. **Firewalls**:
    - Protects cloud systems by filtering incoming and outgoing traffic.
  3. **Backup and Recovery**:a
    - Regularly saving data copies to restore in case of loss.
  4. **Multi-Factor Authentication (MFA)**:
    1. Adds extra layer of security beyond passwords.
  5. **Access Control**:
    1. Restricts who can view or edit data.

## 4.3 Data Security Technologies and Risks

- Data Security Risks:

- **Data Breach:** Unauthorized access to sensitive information.
- **Data Loss:** Deletion or corruption of data due to human error or system failure.
- **Insecure APIs:** Poorly protected interfaces may be exploited by attackers.
- **Insider Threats:** Employees or trusted users misusing access.
- **Lack of Encryption:** If data is not encrypted, it's easy to steal or read.



## 4.4 Digital Identity and Access Management (IAM)

- Define : (i) Cloud security (ii) Digital Identity (4marks)
- Explain digital identity and access management in cloud security. (4M)
- Explain how Identity management can be implement using open stack. (4m) s22

## 4.4 Digital Identity and Access Management (IAM)

- Definition: Digital Identity and Access Management (IAM) refers to the framework of policies, technologies, and processes used to manage digital identities and control access to cloud resources securely.
- It ensures that **only authorized users** can access specific systems or data based on their identity and roles.
- **Example**:
- In a cloud-based hospital system, a **doctor** can access patient records, but a **receptionist** can only view appointment schedules — IAM ensures such controlled access.

# Key Functions of IAM:

- **Authentication:**

- Verifies the identity of users (e.g., username + password, OTP, biometrics).

- **Authorization:**

- Grants or denies access to cloud resources based on user roles and policies.

- **User Identity Management:**

- Maintains user information like name, role, access level, etc.

- **Role-Based Access Control (RBAC):**

- Access is given based on assigned roles (e.g., admin, editor, viewer).

- **Single Sign-On (SSO):**

- Allows users to log in once and access multiple applications without logging in again.

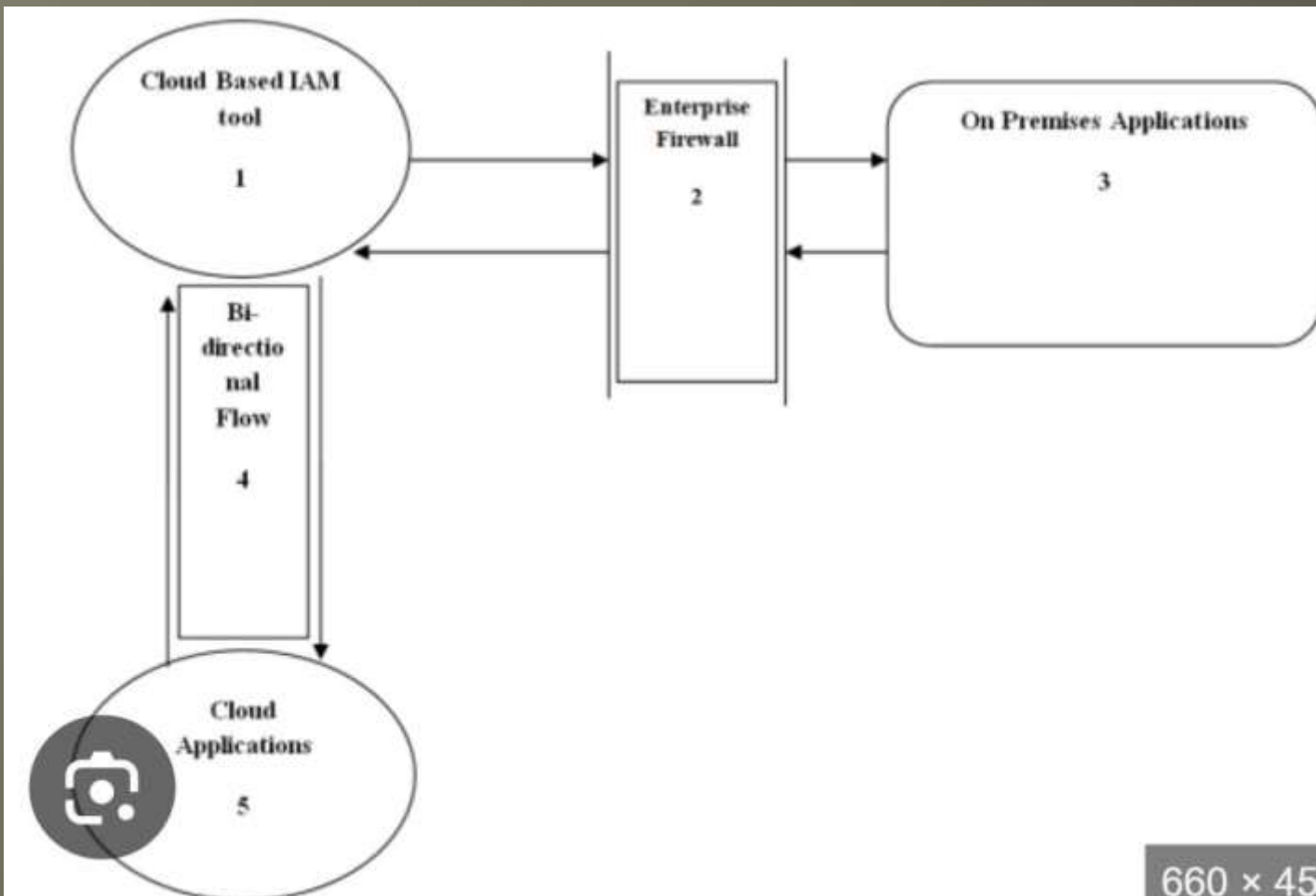
- **Audit and Monitoring:**

- Tracks user activity and detects unauthorized access or misuse.

# Benefits of IAM:

- Prevents unauthorized access to data and resources
- Enhances security and user accountability
- Simplifies user access management
- Supports compliance with security regulations
- Reduces insider threats and data breaches

# Diagram of IAM



## How It Works:

1. When a user requests access, the **Cloud-Based IAM Tool** verifies their identity.
2. If valid, it allows access through the **Firewall** to on-premises or cloud applications.
3. **Bi-directional flow** ensures real-time syncing of user roles, policies, and changes.

## 4.5 Content level security: Pros and Cons, Features of Security-As-A-Cloud Service

- Questions

1. Justify security-as-a-cloud service. (4m) s-24
2. Explain content level security. (4m) s-23
3. Explain content level security in cloud. (4m) w-22
4. Describe content level security. (4m) s-22

Expected : Pros and Cons of Content level security (2m)

Features of Security-As-A-Cloud Service (2m)

# Content Level Security in Cloud Computing

- Content Level Security (CLS) in cloud computing refers to the measures and protocols used to protect data and information at the most granular level
- Content Level Security refers to the protection of **individual data items or files** stored in the cloud by applying security directly on the content (data), regardless of where it is stored or how it is transmitted.



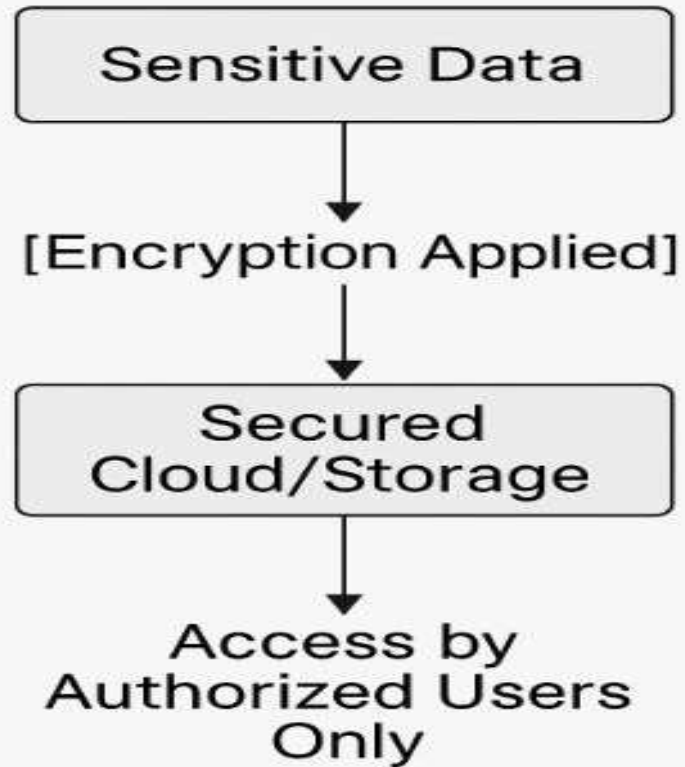
# Explanation .....

- It ensures that the **actual data (content)** is secure, even if attackers bypass network or storage security.
- Security measures like **encryption, digital signatures, and access control** are applied directly on files, emails, documents, or messages.
- This allows **data to stay secure at rest, in transit, and in use.**

## Example:

- An organization encrypts confidential documents stored on Google Drive. Even if the file is downloaded illegally, it cannot be read without the decryption key — this is content-level security.

# Diagram



# Just for knowledge (Not In syllabus)

## Techniques Used:

- **Data Encryption:** Converts data into unreadable format unless decrypted by authorized user.
- **Access Control:** Restricts who can view or edit the content.
- **Digital Rights Management (DRM):** Controls how content is used (e.g., preventing printing or copying).
- **Content Tagging & Classification:** Labels sensitive content to apply appropriate protection

# Advantages /Pros of Content Level Security

1. **End-to-End Data Protection:** Content remains secure during storage, transfer, and usage. Even if the network or storage is compromised, the data stays encrypted.
2. **User-Level Access Control:** Allows defining who can view, edit, or download specific content. Helps in maintaining data confidentiality.
3. **Secure Data Sharing :** Enables secure sharing of documents or files across networks or cloud. Prevents misuse or leakage of sensitive information
4. **Supports Compliance Requirements & Fine-Grained Access Control**

# Disadvantages (Cons):

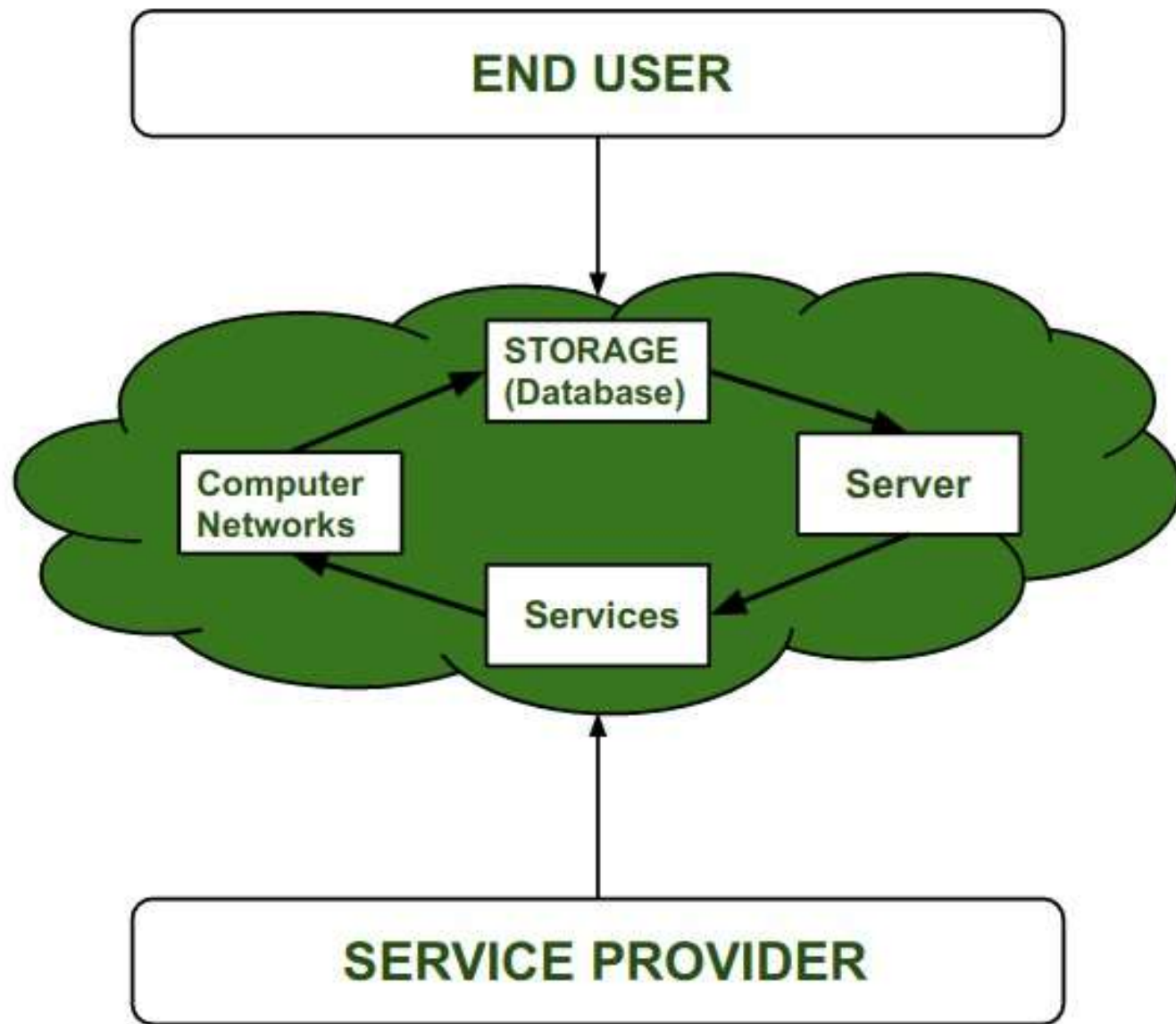
- **Performance Impact:**
  - Encrypting and decrypting content adds processing time.
  - May slow down applications, especially for large files.
- **Complex Key Management:**
  - Managing and securing encryption keys is difficult.
  - If keys are lost or stolen, the data becomes inaccessible.

# Security-As-A-Cloud Service.

- Security as a Service (SECaaS) refers to outsourcing cybersecurity to a third-party provider who delivers security services over the internet.
- This model allows businesses to leverage specialized security expertise and tools without the need for substantial upfront investments in hardware and personnel.
- Essentially, SECaaS is a cloud-based approach to cybersecurity, offering various security solutions like data protection, network security, and more on a subscription basis.
- **Security-as-a-Service (SECaaS)** means the cloud provider delivers security functions (like firewall, antivirus, identity access) as a service over the internet to protect user data, applications, and infrastructure.

## Example:

A company using cloud email security services (like spam filtering, malware detection) without installing anything on its own servers.



# Features of Security-as-a-Service (SECaaS): (2m fixed)

1. **On-demand Security Services:**  
Security tools like antivirus, firewall, intrusion detection, and access control are provided through the cloud whenever needed.
2. **Centralized Management:**  
All security functions can be monitored and managed from a central dashboard.
3. **Scalability:**  
Security services can easily scale up or down as per user or business needs.
4. **Cost-effective:**  
No need to buy expensive hardware or software; pay only for what is used (pay-as-you-go model).
5. **Automatic Updates:**  
Security tools and systems are automatically updated by the provider, reducing the risk of outdated protection.
6. **Data Protection:**  
Provides strong data encryption, secure access, and protection from data breaches.
7. **Real-time Monitoring:**  
Constant surveillance for threats, malware, and unauthorized access attempts.
8. **Compliance Support:**  
Helps businesses meet legal and regulatory security requirements (like GDPR, HIPAA, etc.).



**THANK  
YOU!**

